

GIZMO Operations – Quick Guide

Never lift the safety ground. Working draft: follow SOP004, released grounding drawings, and the current work plan.

Why grounding and regular tests matter

Keep the cryostat and beam structure connected to the required Building Ground during contractor work. Friction and moving materials can generate static charge; grounding drains it and reduces shock and ESD risk. The approved safety-ground path also carries fault current if a powered tool or cable energizes the structure.

Construction can add unwanted ground connections through tools, cables, scaffolding, or metalwork. Regular GIZMO tests find changes in isolation and coupling while they are still accessible to correct, protecting detector noise performance. **GIZMO does not replace the safety bond check.**

Contractor shift checks

At shift start and handover:

1. Inspect grounding cables and connections. Have the designated trained checker verify the Building Ground bond by the approved method.
2. Confirm the grounding connections and tagout status against the work plan; only authorized personnel may change them.
3. Record the check. Clear tools and cords that could bridge the grounds for the agreed FNAL test window.

FNAL I&I/GIZMO isolation tests

Test at the agreed recurring shift windows and after work that could alter grounding or coupling.

1. Coordinate with the contractor. Establish and record the authorized test state. Keep the approved safety-ground path intact.
2. After settling, check fresh, good-quality data, calibration, and frequency. Compare isolation, valid capacitance, coupling, and alarm state with approved limits and the previous equivalent test.
3. Restore and verify the required Building Ground connection; close out tagout and record handback before work resumes.

Log time, checker, grounding state, readings, pass/fail, action, handback, and next test in Procore Daily Log > GIZMO. Attach measurement plot captures to the Procore entry.

When a check fails

- **Safety bond damaged, missing, or unverified:** stop affected work, keep clear, and notify the electrical lead. Resume after defects are corrected, the bond is verified, and handback is authorized.
- **Unexpected alarm or failed GIZMO test:** preserve grounding; notify FNAL I&I/GIZMO and the electrical lead. Investigate and retest under the work plan. Stop affected work if safety is uncertain.
- **Missing, stale, or uncertain data; missed test:** log the gap. FNAL I&I/GIZMO must provide a valid local measurement. If the required check cannot be completed, refer work authorization to the responsible lead.

HIGH Z is not an automatic pass. A dark beacon does not prove safe grounding. Never remove a safety bond or change thresholds to clear an alarm.

Before use: the work plan must name the responsible people and define test methods, intervals, settling time, limits, local fallback, and handback authority. See the extended guide for settings, server checks, and shutdown.